Misalkan variabel random X dan Y mempunyai fungsi densitas bersama

$$f(x,y) = \begin{cases} Cxy, & 0 < y < x < 1 \\ 0, & \text{lainnya} \end{cases}$$

1. Nilai C

Agar $f(x,y)$ merupakan fungsi densitas bersama, haruslah

$$\int_0^1 \int_0^x Cxy \, dy \, dx = 1.$$

Maka

$$\begin{aligned} \int_0^1 \int_0^x Cxy \, dy \, dx &= C \int_0^1 x \left[\frac{y^2}{2} \right]_0^x dx = C \int_0^1 \frac{x^3}{2} \, dx \\ &= \frac{C}{2} \left[\frac{x^4}{4} \right]_0^1 = \frac{C}{8} = 1. \end{aligned}$$

Jadi,

$$\boxed{C = 8}$$

2. Densitas Marginal $f(x)$

$$f(x) = \int_0^x 8xy \, dy = 8x \left[\frac{y^2}{2} \right]_0^x = 4x^3$$

sehingga

$$f(x) = \begin{cases} 4x^3, & 0 < x < 1 \\ 0, & \text{lainnya} \end{cases}$$

3. Fungsi Densitas Bersyarat $f(y \mid x)$

$$f(y \mid x) = \frac{f(x,y)}{f(x)} = \frac{8xy}{4x^3} = \frac{2y}{x^2}$$

sehingga untuk $0 < x < 1$,

$$f(y \mid x) = \begin{cases} \dfrac{2y}{x^2}, & 0 < y < x \\ 0, & \text{lainnya} \end{cases}$$

4. Nilai Harapan Bersyarat $E\left(Y \mid X = \frac{1}{3}\right)$

$$E(Y \mid X = x) = \int_0^x y \cdot \frac{2y}{x^2}\, dy = \frac{2}{x^2}\left[\frac{y^3}{3}\right]_0^x = \frac{2}{x^2} \cdot \frac{x^3}{3} = \frac{2x}{3}.$$

Untuk $x = \frac{1}{3}$:

$$E\left(Y \mid X = \tfrac{1}{3}\right) = \frac{2}{3} \cdot \frac{1}{3} = \boxed{\frac{2}{9}}$$